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# Palaeomagnetic study of some Neogene magmatic rocks from the Oaş–Igniş–Văratec–Tibleş Mountains (Romania)

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## SUMMARY

For this palaeomagnetic study 199 sites were sampled in the northern part of the East Carpathians (Oaş–Igniş–Văratec–Tibleş sector). After AF (Alternating Field) demagnetization, 155 sites were selected for further analyses. A good agreement between the measured palaeomagnetic poles for the Pontian, the Pannonian and the reference polar wander path of Eurasia was made evident. In the southern part of the Igniş Mountains the Sarmatian volcanic rocks were affected by a 30° counterclockwise rotation with respect to Europe post-emplacement.

**Key words:** East Carpathians, Neogene magmatic rocks, palaeomagnetism, Romania.

## 1 INTRODUCTION

On the territory of Romania a strong volcanic activity took place during the Neogene Period. The presence of these Neogene magmatic rocks on the territory of Romania is connected with the Neogene–Quaternary volcanism, with subduction origin, from the Carpatho–Pannonian domain. The calk–alkaline volcanites of this domain are close to the island arc magmatites in their general petrochemistry (Pavelescu 1980; Balla 1985). The volcanic areas are in the inner side of the East Carpathians and in the Apuseni Mountains (Fig. 1). In the East Carpathians the Neogene volcanism is well developed in the northwestern part (Oaş, Igniş, Văratec and Tibleş Mountains = OIVT sector) and in the southern part (Călimani, Gurghiu and Harghita mountains = CGH sector). Between these sectors the neogene volcanic rocks appear only sporadically.

The previous palaeomagnetic studies were performed in the CGH sector (Pătraşcu 1976a). The results obtained from 85 sites yielded mean palaeomagnetic directions with $D = 6°$ $I = 62°$ $\alpha_{95} = 3°$ and $k = 29$, and allowed the separation of this sector into six areas according to the magnetic polarity. The emplacement of the studied volcanic rocks took place between 7 Ma and 3 Ma (Rădulescu, Pătraşcu & Bellon 1972). The palaeomagnetic data are consistence with the absence of tectonic movements with respect to stable Europe post-emplacement.

In this study the palaeomagnetic results from OIVT sector will be reported.

## 2 GEOLOGICAL SETTING AND SAMPLING

Complex geological and geophysical studies have shown that the structural arrangement of the pre-volcanic basement played an important role in the emplacement of the volcanic rocks in OIVT sector (Borcoş *et al.* 1979). The system of crustal faults of the pre-volcanic basement is represented in Fig. 2(a) after Săndulescu (1984). This system was probably tectonically active until Sarmatian and it is covered by the volcanic rocks and coeval sedimentary deposits.

The volcanic activity took place during Late Miocene–Pliocene Period and had three main stages: (1) rhyolite–dacite tuffs and lava flows; (2) a succession of pyroxene-andesite and quartz-andesite strato volcanoes and lava flows; (3) pyroxene andesites and basaltic andesites forming strato volcanoes, which reach the collapse caldera stage (Giuşcă *et al.* 1973). The age assigned for the volcanic rocks is based on the correlations with the sedimentary deposits of the surrounding basins. The ages of the sediments from the basins were determined according to the characteristic fauna for the Pannonian basin. In the sediments imbricated with the volcanic rocks, the fossils are rare, so the correlations with the sediments of the basins are based mainly on the facies similitude. Inside the volcanic areas when the sediments are absent the petrographic characteristics and the relative positions of the volcanic rocks were taken into account to assign the age. A complete review about the age of the volcanic rocks from OIVT sector was made by Borcoş *et al.* (1973).

For this study the Sarmatian, Pannonian and Pontian volcanic rocks of stages 2 and 3 were sampled. This local stratigraphic classification corresponds to the Tortonian and Messinian stages of the Late Miocene Period (Van Eysinga 1975).

650 block samples were collected from 199 sites in OIVT sector (Fig. 2). The block samples were oriented with a solar compass and in absence of sun, with a magnetic compass and a declination gradiometer (Doell & Cox 1962) to avoid the rocks subjected to lightning discharges. The density of the sites was imposed by the number of outcrops and the



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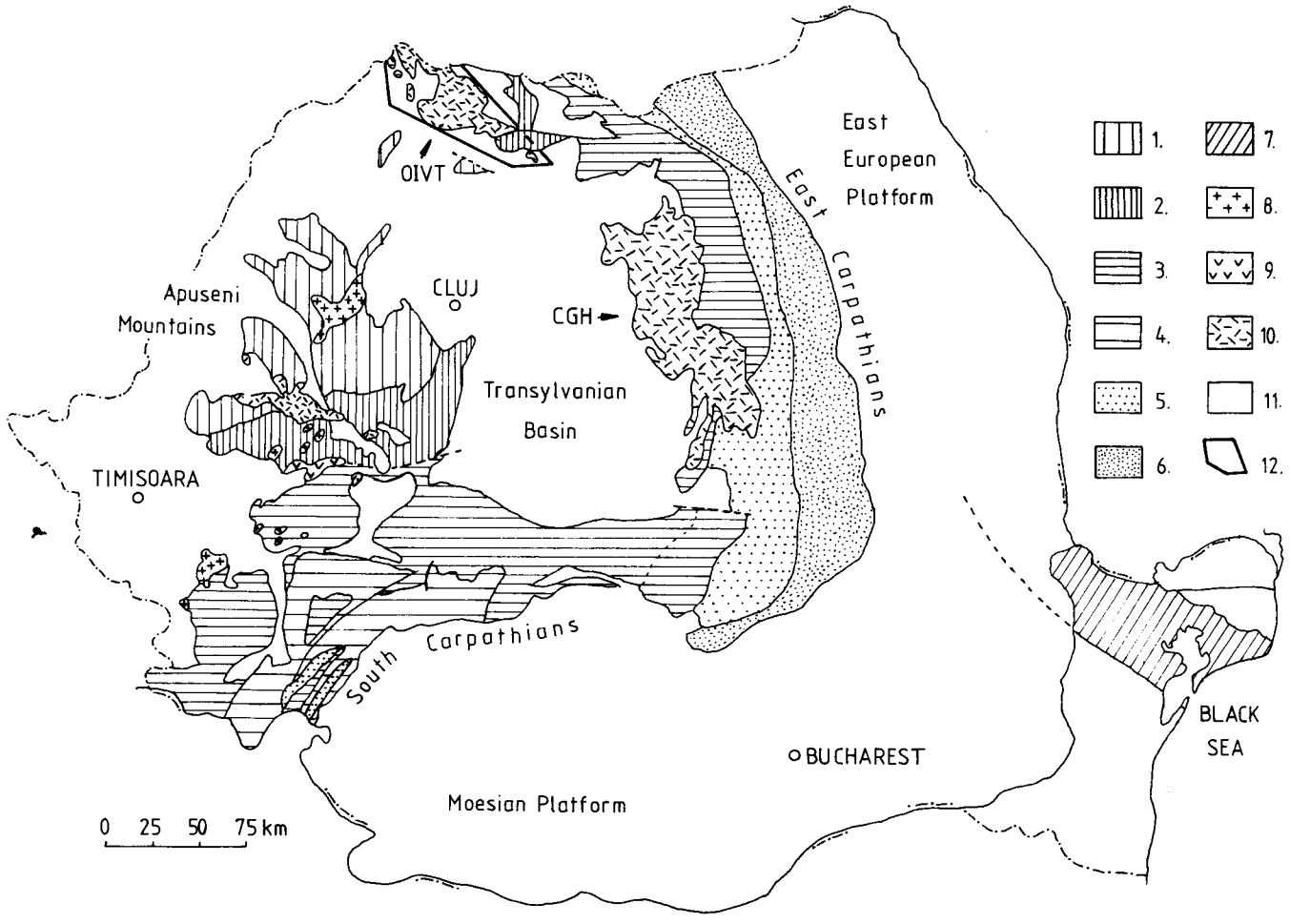



**Figure 1.** Schematic structural map of Romanian territory, after Săndulescu 1984. Symbols: 1 = Internal Dacides, 2 = Transylvanides and Pienides, 3 = Middle Dacides, 4 = Marginal Dacides, 5 = External Dacides, 6 = Moldavides, 7 = Northern Dobrogea Unit, 8 = Upper Cretaceous magmatites, 9 = Oligocene magmatites, 10 = Neogene volcanites, 11 = Post-tectonic cover, 12 = Sampling area. Area symbols: CGH = Călimani–Gurghiu–Harghita Mountains; OIVT = Oaş–Igniş–Văratec–Tibleş Mountains.

quality of rocks. For this study only fresh rocks were retained. The sampled areas in the Oaş, Igniş, Văratec and Tibleş Mountains are not disturbed by a post-volcanic regional system of faults or folds, but local tectonic movements owing to successive intrusions are possible. In the field such displacements were not visible at the sampling sites and no tectonic corrections were applied.

## 3 NRM VECTOR AND MAGNETIC SUSCEPTIBILITY

In the laboratory, four cubic specimens were cut from each sample. The natural remanent magnetization (NRM) vector and the anisotropy of magnetic susceptibility of each specimen were measured using As's method with 24 positions (As 1967) with an astatic magnetometer MA-21. The intensity of NRM are represented in the histogram of Fig. 3(a). Mostly of intensities are between 100 mA $m^{-1}$ and 1000 mA $m^{-1}$. The mean susceptibility is around $10^{-2}$ (SI units) for the majority of specimens (Fig. 3b). The anisotropy of susceptibility is always less than 5 per cent, mostly of specimens having the anisotropy around 1–3 per cent, so their anisotropy is in a range suitable for palaeomagnetic studies (Tarling 1971).

## 4 AF DEMAGNETIZATION OF SPECIMENS

In the laboratory, the structure of the NRM was analysed by alternating field (AF) demagnetizations. The AF demagnetizer was a two-axes tumbler, placed within the field-free space provided by a Helmholtz coils system, capable of maintaining a field of 10 nT or less (Pătraşcu 1976b). In the first stage a pilot specimen of each sample was demagnetized in 10–12 steps up to 100 mT and demagnetization data were plotted using Zijderveld's method (1967). In the second stage, the other specimens from each sample, with well defined directions for the components of the NRM were partially demagnetized to isolate the characteristic magnetization of the sample. In all cases the field used for that was greater than 20 mT.

417 pilot specimens selected from 199 sampling sites were studied, but only for 377 specimens from 155 sampling sites were well defined NRM structures identified. According to the number of linear segments identified on Zijderveld diagrams the following structures were identified: one component (26 specimens, Fig. 4); two component (337 specimens, Fig. 5); three component (14 specimens, Fig. 6).

The scatter of directions of soft and intermediate components between several samples from a single site is

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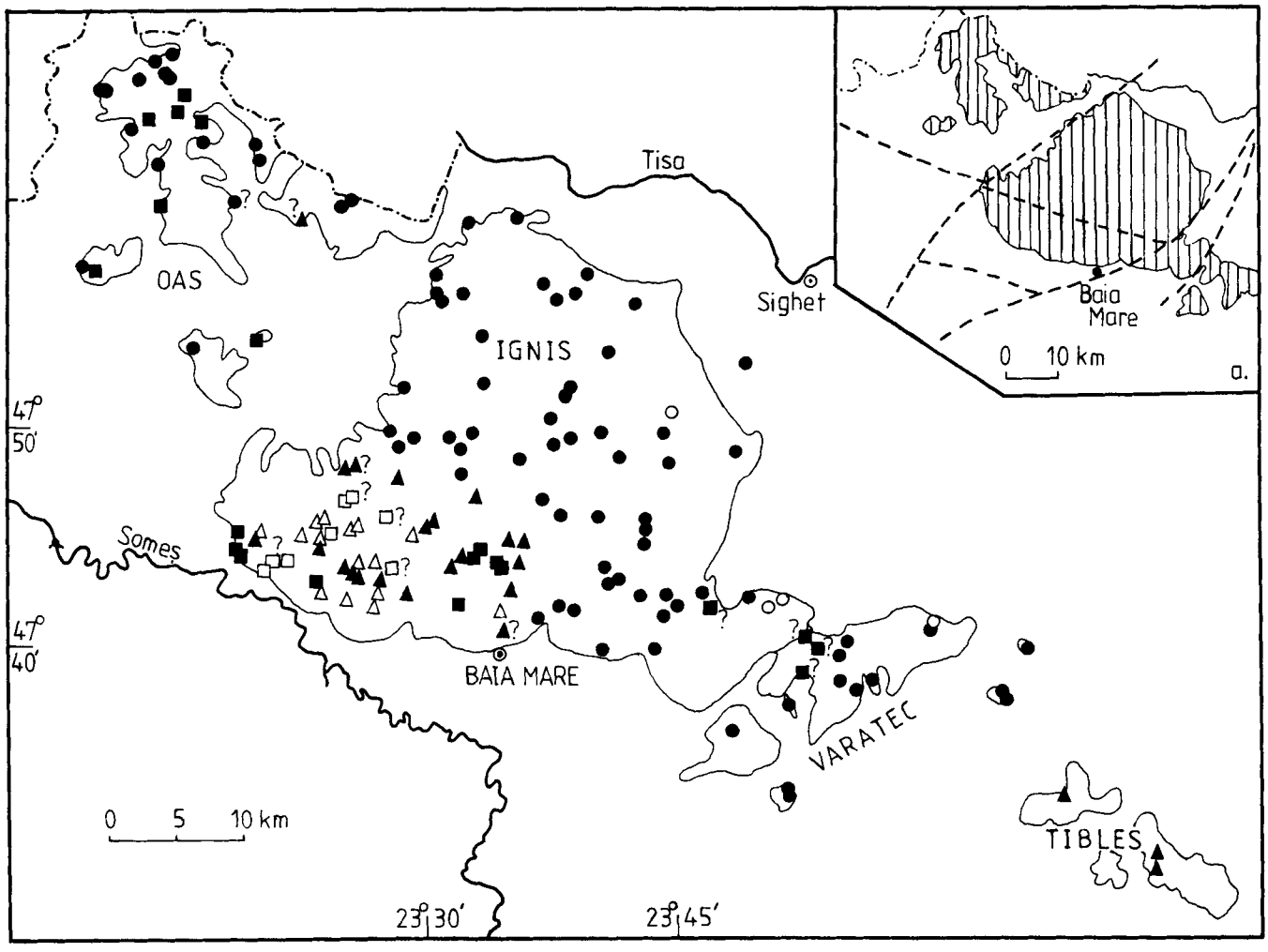



**Figure 2.** Sampling area. Full symbols = sites with positive inclinations, open symbols = sites with negative inclinations; triangle = Sarmatian rocks, square = Pannonian rocks and circle Pontian rocks; ? = rejected sites. (a) Tectonic sketch map of studied area (hatched zones).

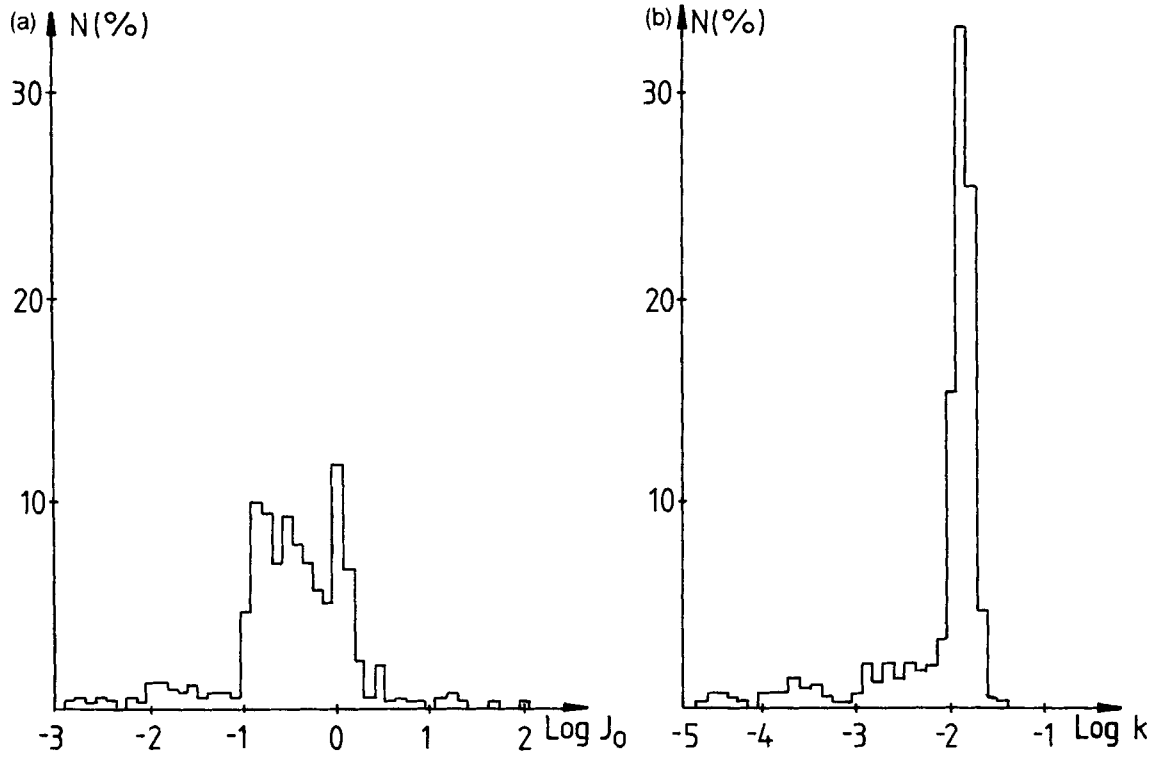



**Figure 3.** (a) Intensity histogram (A $m^{-1}$); (b) Susceptibility histogram (S.I. units); N(%) = percentage of samples.

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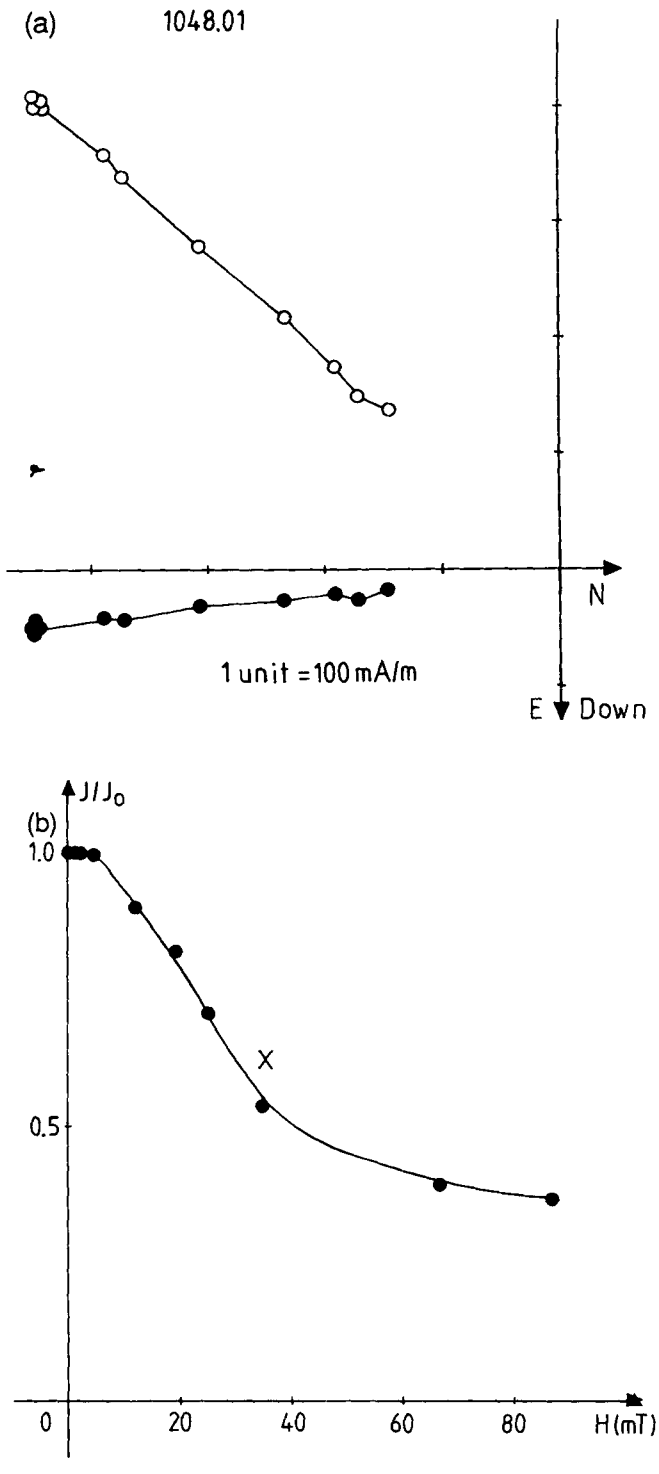



**Figure 4.** Typical example for one-component NRM: (a) Zijderveld diagram, open circle = vertical projection, full circle = horizontal projection. (b) Decay curve, $X$ = hard component.

large and taking into account that these components are removed, mostly around 20 mT, I think that they are probably laboratory viscous remanent magnetizations. In a few cases their directions are aligned with the direction of the present geomagnetic field and then they are probably field viscous remanent magnetizations.

The hard components of the NRM both from

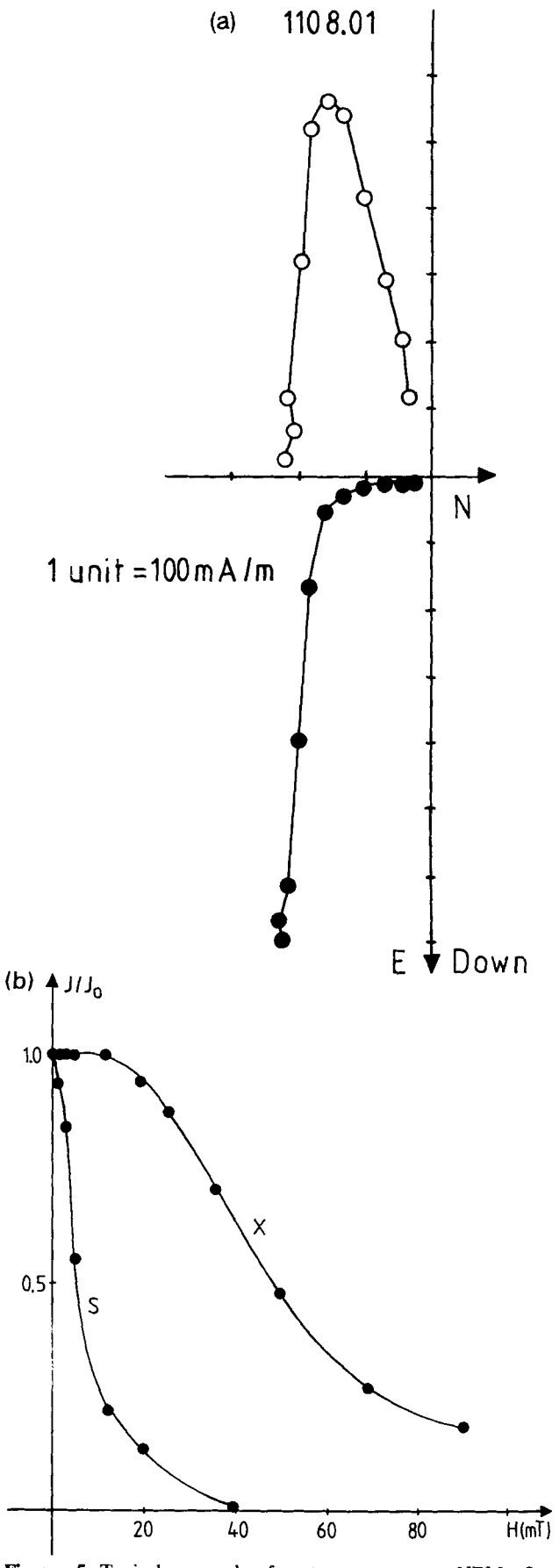



**Figure 5.** Typical example for two-component NRM. Same conventions as Fig. 4. S = soft component, $X$ = hard component.

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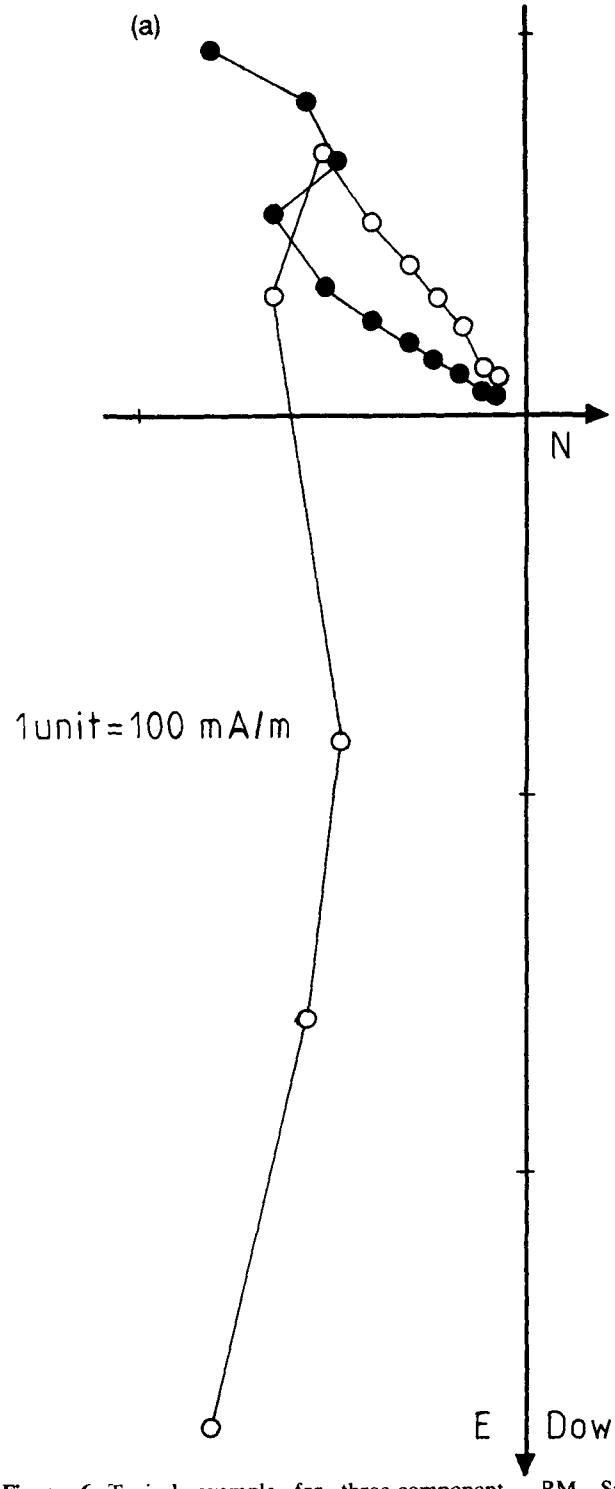



**Figure 6.** Typical example for three-component RM. Same conventions as Fig. 4. S1 = soft component, S2 = intermediate component, $X$ = hard component.

multicomponent and one-component specimens were combined to obtained the site mean directions (Table 1) and have the following characteristics: (1) the within-site scatter is low (Table 1); (2) their decay curves are similar to those obtained for thermo-remanent magnetizations of synthetic samples with magnetite (Dunlop 1981); (3) both normal and reversed polarities were identified with a strong bias toward normal polarity (83 per cent of samples); (4) they can be

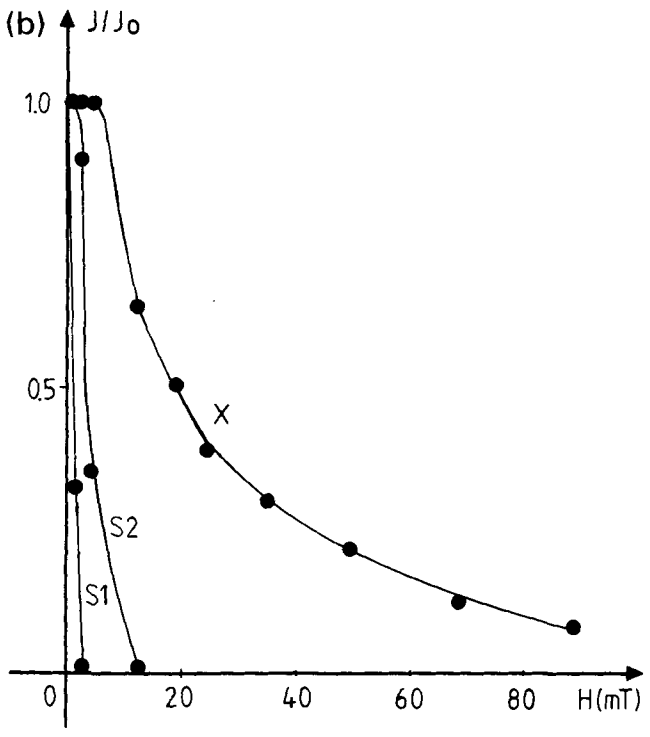



**Figure 6.** (continued)

defined like characteristic magnetizations for relatively large areas.

Mineralogical studies in reflected and transmitted light revealed that the predominant magnetic mineral in the

**Table 1.** Site-mean palaeomagnetic data from studied area.

| Age | Site | Lithology | | D (°) | I (°) | $\alpha_{95}$ (°) | N | k | $\phi$ (°) | $\lambda$ (°) |
|---|---|---|---|---|---|---|---|---|---|---|
| OAS | | | | | | | | | | |
| Po | P25 | andesite(Q) | body | 15 | 60 | 4 | 16 | 75 | 77 | 144 |
| Po | P27 | dacite | body | ? | 64 | | 12 | 185 | 87 | 174 |
| Po | P28 | dacite | body | 14 | 57 | _ | 8 | 1459 | 75 | 153 |
| Po?* | P29 | hyalodacite | flow | 78 | 60 | 2 | 12 | 498 | 37 | 90 |
| Po | P39 | andesite(px) | flow | -1 | 52 | 3 | 12 | 169 | 75 | -154 |
| Po | P40 | andesite(px) | flow | 9 | 56 | 2 | 8 | 737 | 77 | 171 |
| Po | P53 | andesite(px) | body | 28 | 55 | ? | 16 | 363 | 66 | 134 |
| Po | P54 | andesite(px) | flow | 7 | 71 | 5 | 12 | 87 | 82 | 54 |
| Po | P63 | andesite(px) | flow | -2 | 41 | 4 | 16 | 108 | 65 | -152 |
| Po | P64A | andesite(px) | dike | 62 | 74 | 7 | 8 | 65 | 53 | 72 |
| Po | P64B | andesite(px) | dike | 15 | 66 | 4 | 20 | 72 | 80 | 104 |
| Po | P108 | andesite(px) | neck | 66 | 72 | 2 | 12 | 625 | 51 | 75 |
| Po | P109 | andesite(px) | flow | 37 | 70 | 2 | 12 | 467 | 66 | 85 |
| Po | P110 | andesite(px) | flow | 21 | 75 | | 16 | 453 | 72 | 58 |
| Po | P111 | andesite(px) | body | 41 | 51 | ? | 12 | 493 | 55 | 127 |
| Po | P115 | andesite(px) | body | 19 | 59 | 2 | 16 | 250 | 74 | 139 |
| Po | P162 | andesite(px) | flow | -8 | 75 | 4 | 16 | 72 | 76 | 7 |
| Pn | P26 | hyalodacite | flow | -31 | 45 | 8 | 12 | 30 | 57 | -98 |
| Pn | P51 | hyalodacite | flow | 13 | 59 | 3 | 12 | 177 | 78 | 151 |
| Pn | P52 | hyalodacite | flow | -9 | 81 | 3 | 16 | 219 | 66 | 16 |
| Pn | P65 | hyalodacite | flow | -6 | 68 | 3 | 12 | 234 | 85 | -23 |
| Pn | P107 | andesite(px) | flow | 10 | 69 | 3 | 12 | 188 | 82 | 80 |
| Pn | P112 | hyalodacite | flow | -21 | 65 | ? | 12 | 497 | 76 | -64 |
| Pn | P114 | andesite(px) | flow | -17 | 66 | 1 | 12 | 1380 | 79 | -59 |
| Sm?* | P41 | andesite(Q) | flow | 95 | 56 | 4 | 12 | 138 | 23 | 84 |
| IGNIS | | | | | | | | | | |
| Po | P2 | andesite(Q) | flow | -46 | 75 | 3 | 20 | 148 | 61 | -22 |
| Po | P3 | andesite(Q) | flow | 1 | 78 | 3 | 20 | 154 | 72 | 24 |
| Po | P4 | andesite(px) | flow | -11 | 72 | ? | 20 | 398 | 79 | -11 |
| Po | P9 | andesite(Q) | flow | 12 | 71 | 2 | 28 | 160 | 78 | 62 |
| Po | P11 | andesite(Q) | flow | 23 | 79 | 4 | 20 | 66 | 66 | 44 |
| Po | P12 | andesite(px) | flow | 14 | 72 | 5 | 16 | 52 | 77 | 60 |
| Po | P15 | andesite(px) | flow | 3 | 70 | 1 | 20 | 433 | 83 | 43 |
| Po | P23 | andesite(px) | flow | 14 | 73 | 3 | 20 | 170 | 77 | 57 |
| Po | P42 | andesite(px) | flow | 50 | 76 | 3 | 16 | 180 | 58 | 64 |
| Po | P43 | andesite(px) | flow | -23 | 63 | 2 | 12 | 429 | 73 | -70 |
| Po | P44 | andesite(px) | flow | -31 | 57 | 4 | 12 | 98 | 65 | -80 |
| Po | P45 | andesite(Q) | flow | 14 | 48 | 1 | 20 | 1934 | 68 | 169 |
| Po | P46 | andesite(px) | flow | 14 | 73 | 2 | 12 | 412 | 77 | 59 |
| Po | P47 | andesite(px) | flow | -7 | 71 | | 20 | 168 | 81 | -2 |
| Po | P48 | andesite(px) | flow | 23 | 71 | 1 | 12 | 913 | 74 | 74 |
| Po | P49 | andesite(px) | body | 31 | 57 | 1 | 12 | 1594 | 65 | 127 |

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**Table 1.** (Continued.)

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Po | P50 | andesite(px) | body | 6 | 54 | 2 | 16 | 589 | 76 | 181 |
| Po | P55 | andesite(px) | flow | 23 | 35 | 7 | 12 | 37 | 56 | 163 |
| Po | P57 | andesite(px) | flow | 6 | 58 | 6 | 16 | 41 | 80 | 174 |
| Po | P58 | andesite(px) | body | 20 | 72 | 3 | 12 | 183 | 74 | 68 |
| Po | P59 | andesite(px) | sill | -9 | 75 | 7 | 12 | 38 | 76 | 5 |
| Po | P60 | andesite(px) | sill | 27 | 38 | 3 | 12 | 183 | 56 | 155 |
| Po | P74 | hyalodacite | body | -34 | 65 | 2 | 8 | 666 | 68 | -55 |
| Po | P83 | andesite(Q) | flow | 12 | 51 | 4 | 16 | 67 | 71 | 170 |
| Po | P84 | andesite(Q) | neck | 5 | 43 | 3 | 12 | 212 | 67 | 191 |
| Po | P123 | andesite(px) | body | -33 | 66 | 3 | 12 | 174 | 68 | -54 |
| Po | P124 | andesite(px) | flow | -50 | 77 | 2 | 16 | 386 | 59 | -15 |
| Po | P125 | andesite(px) | flow | -5 | 58 | 3 | 12 | 179 | 81 | -134 |
| Po | P127 | andesite(px) | flow | -7 | 65 | 1 | 12 | 1971 | 85 | -74 |
| Po | P128 | andesite(px) | flow | 2 | 75 | 2 | 12 | 361 | 77 | 28 |
| Po | P129 | andesite(px) | flow | 7 | 63 | 3 | 12 | 280 | 84 | 142 |
| Po | P130 | andesite(px) | flow | 12 | 70 | 2 | 12 | 758 | 81 | 74 |
| Po | P131 | andesite(px) | flow | -11 | 67 | 2 | 12 | 736 | 82 | -47 |
| Po | P139 | andesite(px) | flow | 19 | 70 | 3 | 12 | 190 | 76 | 77 |
| Po | P140 | andesite(px) | dike | -42 | 65 | 2 | 12 | 465 | 62 | -50 |
| Po | P141 | andesite(px) | flow | -22 | 59 | 1 | 12 | 1113 | 72 | -84 |
| Po | P142 | andesite(px) | neck | 39 | 46 | 4 | 16 | 106 | 54 | 133 |
| Po | P143 | andesite(px) | neck | 8 | 63 | 2 | 12 | 362 | 84 | 139 |
| Po | P146 | andesite(px) | flow | -29 | 36 | 3 | 16 | 143 | 54 | -106 |
| Po | P164 | andesite(px) | flow | 28 | 40 | 4 | 12 | 120 | 56 | 152 |
| Po | P165 | andesite(px) | body | 11 | 26 | 5 | 12 | 83 | 55 | 184 |
| Po | P166 | andesite(px) | flow | 16 | 51 | 5 | 12 | 71 | 69 | 161 |
| Po | P167 | andesite(px) | flow | 4 | 54 | 2 | 12 | 310 | 76 | 188 |
| Po | P174 | andesite(px) | flow | -8 | 61 | 4 | 8 | 210 | 81 | -110 |
| Po | P175 | andesite(px) | flow | 15 | 72 | 1 | 12 | 1175 | 76 | 59 |
| Po | P176 | andesite(px) | flow | 48 | 78 | 2 | 12 | 302 | 59 | 58 |
| Po | P179 | andesite(px) | flow | -15 | 70 | 1 | 12 | 965 | 79 | -31 |
| Po | P188 | andesite(px) | flow | 51 | 61 | 2 | 12 | 360 | 54 | 104 |
| Po | P194 | andesite(px) | body | -30 | 76 | 2 | 12 | 743 | 68 | -12 |
| Po | P195 | andesite(px) | body | -17 | 50 | 6 | 12 | 61 | 69 | -112 |
| Po | P196 | andesite(px) | body | -38 | 71 | 3 | 8 | 467 | 66 | -36 |
| Po | P197 | andesite(px) | flow | 0 | 67 | 3 | 12 | 309 | 89 | 14 |
| Po | P198 | andesite(px) | sill | 9 | 46 | 4 | 12 | 145 | 68 | 181 |
| Po | P199 | andesite(px) | flow | 23 | 62 | 5 | 12 | 77 | 73 | 124 |
| Po | P56 | andesite(px) | flow | 154 | -34 | 7 | 12 | 36 | -54 | 69 |
| Po | P89 | andesite(px) | flow | -161 | -61 | 2 | 12 | 505 | -75 | -51 |
| Po | P173 | andesite(px) | flow | -172 | -68 | 5 | 12 | 90 | -84 | -104 |
| Pn | P1 | andesite(px) | flow | 20 | 69 | 1 | 20 | 775 | 76 | 87 |
| Pn | P21 | andesite(Q) | flow | 24 | 66 | 2 | 20 | 330 | 74 | 101 |
| Pn?* | P62 | andesite(Q) | flow | 63 | 70 | 3 | 16 | 149 | 51 | 80 |
| Pn | P87 | andesite(Q) | flow | 39 | 56 | 2 | 8 | 1251 | 59 | 121 |
| Pn | P97 | andesite(px) | flow | -3 | 69 | 2 | 12 | 417 | 85 | 0 |
| Pn | P161 | andesite(px) | flow | 1 | 62 | 1 | 12 | 913 | 85 | 195 |
| Pn | P190 | dacite | flow | 35 | 61 | 1 | 8 | 1981 | 65 | 113 |
| Pn | P191 | dacite | flow | 21 | 53 | 7 | 12 | 40 | 69 | 147 |
| Pn | P192 | andesite(Q) | flow | 42 | 61 | 3 | 4 | 707 | 60 | 108 |
| Pn | P193 | andesite(Q) | flow | 45 | 61 | 2 | 12 | 509 | 58 | 106 |
| Pn | P5 | andesite(px) | flow | -172 | -61 | 1 | 20 | 576 | -82 | -24 |
| Pn | P73 | andesite(px) | flow | -161 | -46 | 4 | 16 | 76 | -65 | -20 |
| Pn?* | P81 | andesite(px) | body | -157 | -45 | 5 | 16 | 68 | -62 | -25 |
| Pn | *P91 | andesite(px) | flow | -136 | -46 | 7 | 24 | 19 | -51 | -51 |
| Pn | P151 | andesite(px) | flow | -165 | -48 | 7 | 12 | 36 | -68 | -13 |
| Pn | *P152 | andesite(px) | flow | -128 | -57 | 2 | 12 | 320 | -51 | -71 |
| Pn | P153 | andesite(px) | flow | -158 | -49 | 3 | 8 | 267 | -66 | -28 |
| Pn?* | P186 | andesite(px) | flow | -152 | -43 | 3 | 12 | 175 | -59 | -30 |
| Sm | P19 | andesite(px) | body | -17 | 59 | 5 | 20 | 38 | 76 | -92 |
| Sm | P20 | andesite(px) | body | 3 | 30 | 4 | 20 | 74 | 58 | 198 |
| Sm | P22 | andesite(px) | body | -20 | 54 | 4 | 20 | 70 | 70 | -100 |
| Sm | P30 | andesite(px) | dike | 4 | 63 | 7 | 20 | 20 | 85 | 163 |
| Sm | P32 | andesite(px) | flow | -44 | 56 | 3 | 24 | 124 | 56 | -70 |
| Sm | *P76 | andesite(px) | flow | -62 | 61 | 2 | 12 | 509 | 47 | -49 |
| Sm | P77 | andesite(px) | flow | -53 | 63 | 2 | 12 | 415 | 54 | -51 |
| Sm | P82A | andesite(px) | flow | -34 | 55 | 4 | 24 | 53 | 62 | -79 |
| Sm | P88 | andesite(Q) | flow | -45 | 61 | 5 | 8 | 116 | 58 | -59 |
| Sm | P98 | andesite(px) | flow | -21 | 61 | 4 | 16 | 76 | 74 | -79 |
| Sm | P117 | andesite(Q) | flow | -13 | 45 | 2 | 12 | 291 | 67 | -126 |
| Sm | *P119 | andesite(Q) | flow | -71 | 53 | 3 | 12 | 225 | 36 | -54 |
| Sm | P120 | andesite(px) | dike | -5 | 54 | 3 | 12 | 234 | 76 | -140 |
| Sm | P121 | andesite(px) | flow | -24 | 71 | 2 | 12 | 430 | 73 | -32 |
| Sm | P148 | andesite(px) | flow | -21 | 73 | 4 | 12 | 135 | 74 | -19 |
| Sm | P160 | andesite(px) | flow | -46 | 51 | 4 | 12 | 142 | 52 | -74 |
| Sm | P177 | andesite(px) | dike | -44 | 58 | 3 | 12 | 216 | 57 | -65 |
| Sm | P180 | andesite(px) | body | -17 | 68 | 2 | 12 | 558 | 78 | -45 |
| Sm | P181 | andesite(px) | flow | -6 | 71 | 1 | 12 | 1041 | 82 | 0 |
| Sm | P185 | andesite(Q) | flow | -38 | 58 | 2 | 12 | 270 | 61 | -70 |
| Sm | P8 | andesite(px) | flow | 160 | -35 | 1 | 20 | 746 | -57 | 60 |
| Sm | P31 | andesite(px) | flow | 159 | -54 | 5 | 12 | 67 | -69 | 80 |
| Sm | P33 | andesite(px) | flow | 148 | -69 | 5 | 16 | 51 | -69 | 138 |
| Sm | P78 | andesite(px) | flow | 178 | -54 | 4 | 28 | 42 | -77 | 29 |
| Sm | P79 | andesite(px) | flow | 126 | -60 | 1 | 20 | 562 | -52 | 123 |
| Sm | P86 | andesite(px) | flow | 177 | -49 | 5 | 12 | 84 | -72 | 32 |
| Sm | P95 | andesite(px) | flow | -158 | -58 | 5 | 16 | 47 | -72 | -46 |
| Sm | P102 | andesite(px) | flow | 144 | -38 | 6 | 12 | 48 | -51 | 83 |
| Sm | P104 | andesite(px) | flow | 160 | -44 | 5 | 12 | 83 | -63 | 66 |
| Sm | P105 | andesite(px) | flow | 156 | -62 | 3 | 16 | 172 | -73 | 108 |
| Sm | P118 | andesite(px) | flow | 173 | -44 | 3 | 12 | 255 | -67 | 40 |
| Sm | P149 | andesite(px) | flow | 151 | -65 | 2 | 12 | 407 | -70 | 121 |
| Sm | P159 | andesite(px) | flow | 166 | -55 | 3 | 16 | 104 | -74 | 68 |

samples is the primary magnetite (or titano magnetite) and in a few cases deuteric magnetite. Taking into account that all these magnetic phases have the same age as the volcanic rocks and the characteristics of the hard components, I think that they are primary characteristic remanent magnetizations.

**Table 1.** (Continued.)

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Sm | P184 | andesite(px) | flow | -176 | -66 | 2 | 12 | 387 | -87 | -90 |
| Sm | P187 | andesite(px) | flow | -176 | -60 | 3 | 12 | 178 | -83 | -3 |
| VARATEC | | | | | | | | | | |
| Po | P16 | andesite(px) | flow | -23 | 73 | 3 | 20 | 100 | 72 | -17 |
| Po | P17 | andesite(px) | flow | -2 | 42 | 8 | 20 | 17 | 67 | -151 |
| Po | P35 | diorite(px) | body | -11 | 51 | 4 | 20 | 68 | 72 | -124 |
| Po | P36 | andesite(px) | flow | 5 | 70 | 7 | 12 | 42 | 83 | 51 |
| Po | P37 | andesite(px) | flow | 9 | 75 | 2 | 16 | 379 | 75 | 40 |
| Po | P38 | andesite(Q) | flow | 20 | 77 | 7 | 8 | 68 | 70 | 49 |
| Po | P72 | dacite | body | 1 | 70 | 9 | 4 | 98 | 84 | 27 |
| Po | P132 | andesite(px) | flow | -9 | 70 | 2 | 12 | 556 | 82 | -17 |
| Po | P136 | andesite(px) | body | 37 | 66 | 3 | 12 | 177 | 65 | 98 |
| Po | P156 | andesite(px) | flow | -25 | 68 | 4 | 12 | 141 | 73 | -45 |
| Po | P169 | andesite(px) | flow | 11 | 75 | 2 | 12 | 389 | 75 | 45 |
| Po | P170 | andesite(px) | flow | -20 | 75 | 3 | 12 | 184 | 71 | -6 |
| Po | P172 | dacite | body | 8 | 70 | 2 | 16 | 454 | 82 | 61 |
| Po | P157 | andesite(px) | flow | 174 | -63 | 4 | 12 | 102 | -84 | 73 |
| Pn | *P18 | andesite(Q) | flow | -90 | 30 | 5 | 20 | 44 | 12 | -55 |
| Pn | *138 | andesite(px) | flow | -46 | 72 | 3 | 16 | 228 | 61 | -32 |
| Pn | *P145 | andesite(px) | flow | -88 | 55 | 4 | 12 | 100 | 27 | -42 |
| TIBLES | | | | | | | | | | |
| Sm | P66 | andesite(Q) | body | 29 | 67 | 4 | 16 | 68 | 70 | 96 |
| Sm | P67 | andesite(px) | body | -8 | 60 | 5 | 12 | 69 | 81 | -109 |
| Sm | P71 | andesite(px) | body | 22 | 71 | 2 | 12 | 313 | 74 | 77 |

Symbols: *D, I, $\alpha_{95}$, k*: declination, inclination, confidence circle and precision of the site-mean; *N* = number of specimens; $\phi$, $\lambda$: latitude and longitude of VGP. Means, confidence circle and precision parameters are determined using Fisher's statistics (1953). Age symbols: Po = Pontian, Pn = Pannonian, Sm = Sarmatian. Lithology symbols: Q = quartz, px = pyroxene. Other symbols: * = rejected sites, ? = uncertain age.

## 5 PALAEOMAGNETIC RESULTS

The site mean directions and the lithologies for the 155 sites selected after AF demagnetizations are summarized in Table 1 and represented in Fig. 7. They are grouped inside each sampling area (Oaş, Igniş, Văratec and Tibleş) according to their age. Several comments regarding this table can be made.

(1) Several sites, marked with * in Table 1, were omitted from further analyses (reversal test, area mean, tectonic interpretations). Five of them, marked with ? have the age defined in large limits. The others were rejected because the colatitude of their VGPs are more than 40° from the mean palaeomagnetic pole for that age or the between-site scatter is large (the three Pannonian sites from Văratec). Several causes can be invoked to explain these directions: geomagnetic excursions, transition field between two polarities, tectonic disturbances produced by successive volcanic events, incorrect age.

(2) The Sarmatian sites from Igniş have both normal and reversed polarities and the reversal test was positive with classification *B* (McFadden & McElhinny 1990).

(3) The Pannonian sites from Igniş have both normal and reversed polarities, but the reversal test was negative. A negative reversal test was also obtained combining the Pannonian sites from Oaş (only normal polarities) with those from Igniş. Taking into account that the Pannonian sites with reversed polarity are very close to the Sarmatian sites with reversed polarity (Fig. 2) I think that probably some of them were wrongly assigned to the Pannonian and this can be explained by the negative reversal test.

(4) The Pontian sites from Igniş have normal polarity with the exception of three sites. The reversal test was positive with classification 'indeterminate'.

The site-mean directions (Table 1) were combined to compute the area mean for different ages and magnetic

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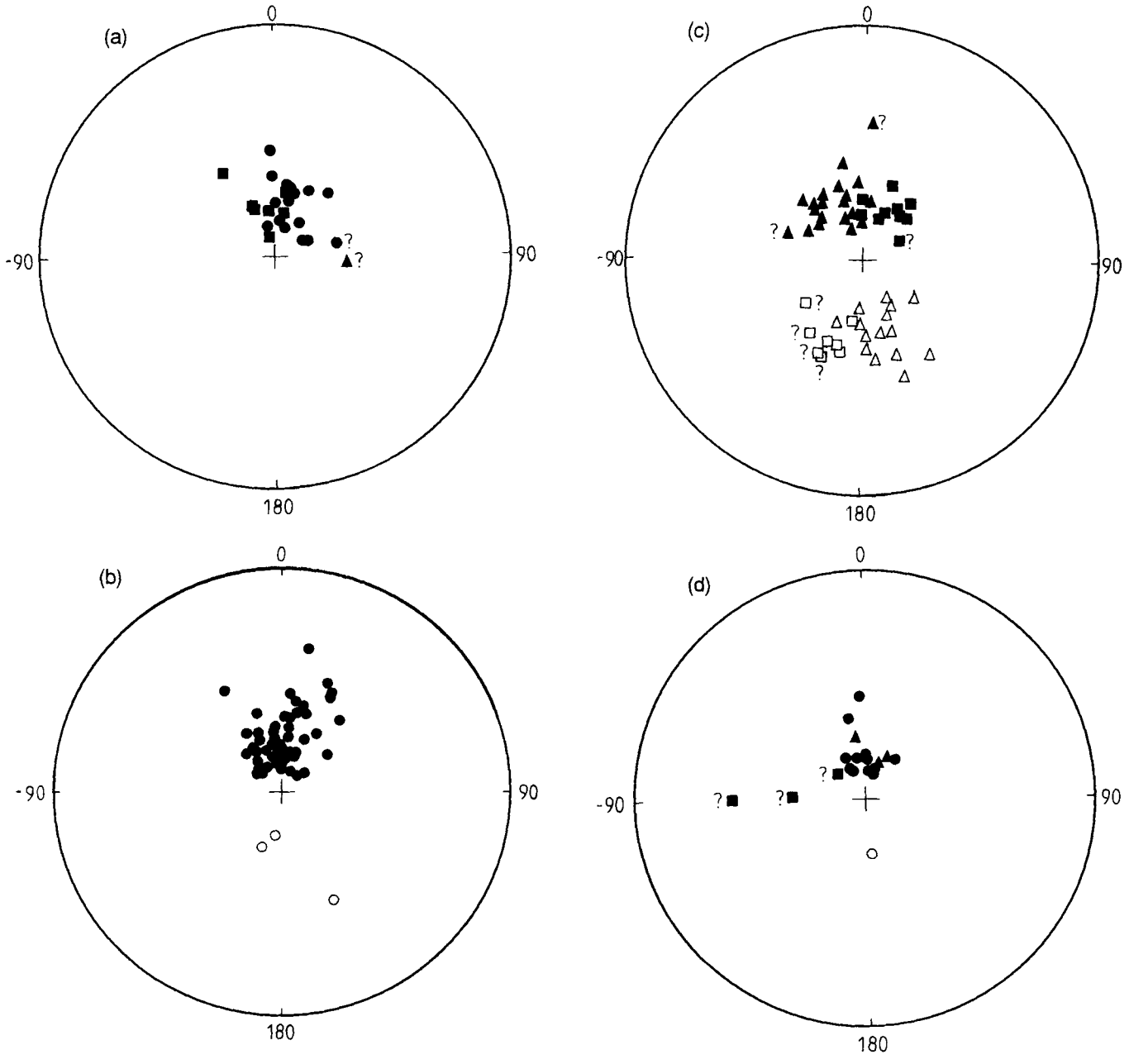



**Figure 7.** Stereographic projection of site-mean directions after AF demagnetization: a = Oaş Mountains, b = Pontian data from Igniş Mountains, c = Pannonian and Sarmatian data from Igniş Mountains, d = Vărateс and Tibleş Mountains. Symbols: triangle = Sarmatian, square = Pannonian, circle = Pontian. Full symbols = site with positive inclinations, open symbol = site with negative inclination; ? = rejected sites.

polarities (lines 1–11, Table 2). In lines 12–18 of Table 2 the means were computed for each age and area combining both normal and reversed polarities. In addition, the Pannonian data from Oaş and Igniş and the Pontian data from Oaş, Igniş and Vărateс were combined to define an area mean for these ages (lines 20 and 19, Table 2).

## 6 TECTONIC INTERPRETATION OF THE DATA

For the tectonic interpretations the palaeomagnetic data from the OIVT sector were compared with the Miocene palaeomagnetic pole from the pure polar wander path for Eurasia (Fig. 8) after Westphal *et al.* (1986). Rotation and flattening parameters (Table 3) were calculated according to the equations of Beck (1980), modified by Demarest (1983).

For the Sarmatian, the palaeomagnetic results were obtained from the southern part of Igniş and Tibleş. Because the Tibleş Mountains were isolated with respect to the volcanic area of Igniş, and the three sites from Tibleş yielded an area-mean direction different from the Sarmatian area-mean direction from Igniş, I did not combine the Sarmatian data from Igniş and Tibleş. Taking into account that the Sarmatian area mean of Tibleş was recovered from only three sites I think that this area mean cannot be used for the tectonic interpretation for the moment. On the

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**Table 2.** Area-mean palaeomagnetic data from OIVT sector.

| Area | Age | P | $N_0$ | D (°) | I (°) | $\alpha_{95}$ (°) | k | $\phi$ (°) | $\lambda$ (°) | $\delta_p$ (°) | $\delta_m$ (°) | $\phi'$ (°) | $\lambda'$ (°) | $A_{95}$ (°) | K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Oaş | Po | N | 16 | 18 | 63 | 6 | 37 | 77 | 119 | 8 | 4 | 77 | 112 | 9 | 20 |
| | Pn | N | 7 | -11 | 65 | 10 | 37 | 83 | -66 | 13 | 7 | 84 | -54 | 13 | 21 |
| Igniş | Po | N | 54 | 4 | 64 | 4 | 25 | 87 | 148 | 5 | 3 | 88 | 102 | 5 | 15 |
| | Po | R | 3 | 174 | -56 | 37 | 12 | -78 | 47 | 38 | 29 | -81 | 36 | 41 | 10 |
| | Pn | N | 9 | 26 | 63 | 6 | 73 | 72 | 114 | 7 | 4 | 72 | 113 | 9 | 35 |
| | Pn | R | 4 | -163 | -51 | 9 | 111 | -70 | -21 | 8 | 7 | -70 | -21 | 9 | 93 |
| | Sm | N | 18 | -24 | 60 | 6 | 34 | 70 | -79 | 7 | 5 | 71 | -76 | 8 | 19 |
| | Sm | R | 15 | 165 | -56 | 7 | 31 | -74 | 74 | 7 | 6 | -75 | 76 | 9 | 21 |
| Văratec | Po | N | 13 | -1 | 69 | 6 | 47 | 85 | 18 | 9 | 4 | 84 | 21 | 8 | 26 |
| | Po | R * | 12 | 174 | -63 | 4 | 102 | -85 | 73 | 5 | 7 | -86 | 72 | 6 | 98 |
| Tibles | Sm | N | 3 | 12 | 67 | 15 | 68 | 82 | 94 | 21 | 25 | 81 | 93 | 23 | 29 |
| Oas | Po | N | 16 | 18 | 63 | 6 | 37 | 77 | 119 | 8 | 4 | 77 | 112 | 9 | 20 |
| | Pn | N | 7 | -11 | 65 | 10 | 37 | 83 | -66 | 13 | 7 | 84 | -54 | 13 | 21 |
| Ignis | Po | N+R | 57 | 3 | 64 | 4 | 24 | 87 | 159 | 5 | 3 | 88 | 116 | 5 | 15 |
| | Pn | N+R | 13 | 22 | 59 | 5 | 58 | 72 | 130 | 6 | 4 | 72 | 127 | 7 | 34 |
| | Sm | N+R | 33 | -19 | 58 | 4 | 33 | 73 | -92 | 5 | 4 | 73 | -88 | 6 | 21 |
| Văratec | Po | N+R | 14 | -1 | 68 | 6 | 50 | 86 | 12 | 8 | 4 | 85 | 18 | 8 | 28 |
| Tibles | Sm | N | 3 | 12 | 67 | 15 | 68 | 82 | 94 | 21 | 25 | 81 | 93 | 23 | 29 |
| Oaş+Ignis+ Varatec | Po | N+R | 87 | 5 | 65 | 3 | 27 | 86 | 131 | 4 | 2 | 86 | 99 | 4 | 16 |
| Oas+Ignis | Pn | N+R | 20 | 12 | 62 | 6 | 34 | 80 | 134 | 7 | 4 | 81 | 129 | 8 | 19 |

Symbols: P = Polarity (N–Normal, R–Reversal); $N_0$ = Number of sites (*$N_0$–specimens from one site); $D$, $I$, $\alpha_{95}$, $k$: declination, inclination, confidence circle and precision of the site-mean; $\phi$, $\lambda$: latitude and longitude of mean pole calculated from the area-mean direction and the 95 per cent semi-axes of the elliptical error $\delta_p$, in colatitude direction, and $\delta_m$, perpendicular to it; $\phi'$, $\lambda'$, $A_{95}K$: latitude and longitude of mean pole calculated from VGPs, confidence circle and precision of the mean. Means, confidence circle and precision parameters are determined using Fisher's statistics (1953). Age symbols according to Table 1.

**Table 3.** Concordance/discordance statistics for OIVT sector.

| Pole | Age | R ± ΔR | F ± ΔF |
|---|---|---|---|
| Ignis | Sarmatian | -30.3°± 9.6° | 1.0°± 6.5° |
| Oaş+Ignis | Pannonian | 2.6°± 11.5° | -0.3°± 6.8° |
| Oas+Ignis+Varatec | Pannonian | -3.8°± 8.7° | -6.5°± 5.4° |

Rotation (R) and flattening (F) parameters calculated according to Beck (1980) and Demarest (1983).

contrary, the Sarmatian area mean from Igniş, based on 33 sites and with a positive reversal test, is significant for a tectonic interpretations. The rotation parameter shows an important counter-clockwise rotation ($-30° \pm 9°$) but no significant latitudinal movements.

The Pannonian palaeomagnetic data were recovered from the southern part of the Igniş Mountains and the Oaş Mountains. Both area-mean directions are computed from a few sites collected in lava flows and are not constrained by a positive reversal test. Therefore, I am not sure that the secular variation is averaged separately for each area and consequently the use of these area means for the tectonic interpretations is questionable. However, taking into account that there is no post-volcanic tectonic argument to separate these areas into independent units, I combined all the Pannonian data in one mean. According to the data of Table 3, the rotation and the flattening parameters are not

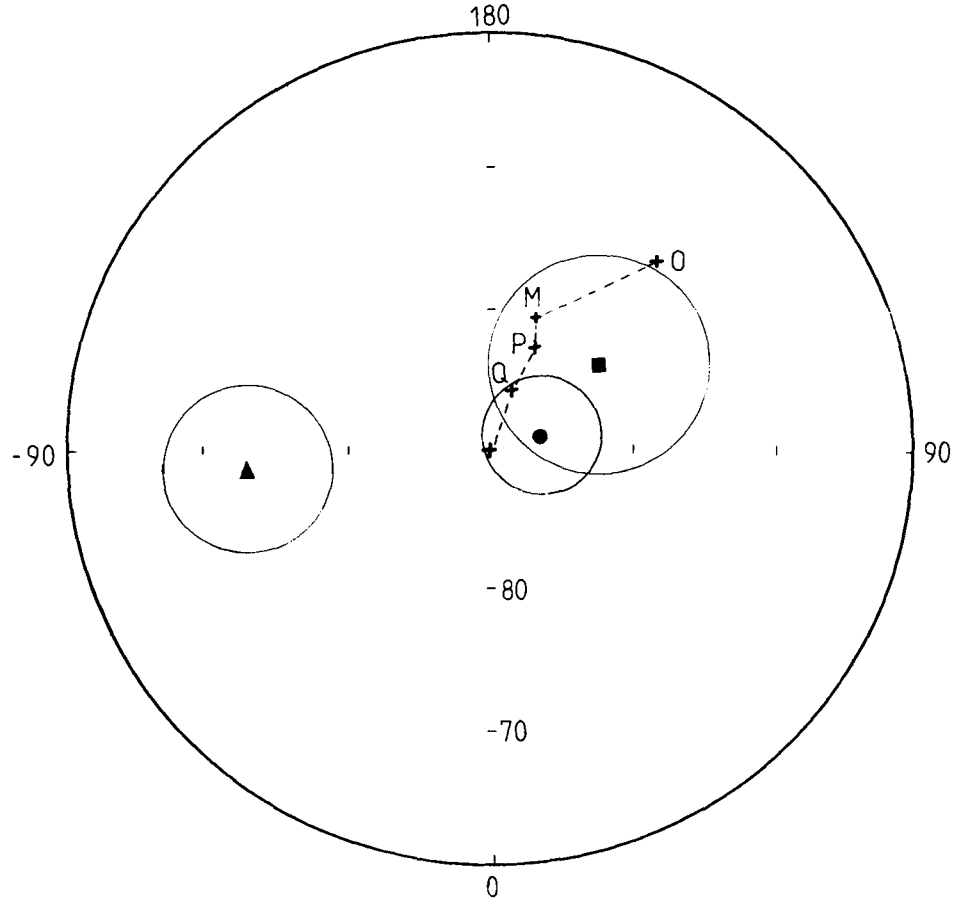



**Figure 8.** Stereographic projection of paleomagnetic poles for sampling area in northern hemisphere. Symbols: triangle = Sarmatian, square = Pannonian, circle = Pontian. Symbols for pure Eurasian reference polar wander pass (cross): Q = Quaternary P = Pliocene, M = Miocene, O = Oligocene (Westphal *et al.* 1986).

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significantly different from zero for this area-mean direction.

Taking into account the same lack of tectonic arguments to separate the Oaş, Igniş and Văratec Mountains into independent units, I combined the Pontian data of these units into one area mean. The rotation parameter is not significantly different from zero, but the flattening parameter is barely significantly different from zero when the reference Miocene pole from the pure Eurasian polar wander curve is used. On the other hand the use of 10 Ma pole from the Synthetic Eurasian polar wander curve (Westphal *et al.* 1986, Table 4) indicates no latitudinal movement ($-3.6° \pm 4.4°$). So the evidence of latitudinal movement of these units with respect to the stable craton depends on the particular choice of the reference pole, and is consequently not really compelling.

## 7 CONCLUSIONS

(1) The present palaeomagnetic results followed the time extension of palaeomagnetic studies in Neogene magmatic rocks of East Carpathians, to older ages than those studied in the Călimani–Gurghiu-Harghita sector (Pătraşcu 1976a).

(2) The volcanic activity in the studied area took place in several polarity intervals with a strong bias toward normal polarity during the Pannonian and Pontian. It must be observed that there is a good agreement between the normal polarity from the Oaş area and the normal polarity obtained by Mihailova, Glevasskaya & Tsykora (1974) in the other parts of the Oaş Mountains, beyond the border of Romania.

(3) The Pontian pole for Oaş, Igniş and Văratec and the Pannonian pole for Oaş and Igniş are in agreement with the Miocene pole from the pure Eurasian polar wander path. So the Pontian and Pannonian volcanic rocks sampled for this study were not affected by regional tectonic movements post-emplacement.

(4) The Sarmatian pole from the Igniş Mountains is significantly different from both younger poles taken from the studied area and the reference Miocene pole of Eurasia, and make evident a counter-clockwise rotation around 30° with respect to a stable Europe.

(5) Previous palaeomagnetic results recovered from the Upper Cretaceous magmatic rocks of the Apuseni Mountains and the South Carpathians (Pătraşcu, Bleahu & Panaiotu 1990; Pătraşcu, Panaiotu & Panaiotu 1991; Pătraşcu *et al.* 1992), from some remagnetized Mesozoic rocks of the northwestern part of the Apuseni Mountains (Surmont, Bordea & Săndulescu 1990) and from the Eocene–Oligocene and Neogene magmatic rocks of the Apuseni Mountains (Pătraşcu *et al.* in preparation) show the existence of a microplate in the eastern part of the Carpatho–Pannonian domain characterized by a clockwise rotation, around 60°, during the Early Neogene. The rotation of the Southern Igniş is the first counter-clockwise rotation identified in the eastern part of this domain. It is probably connected with the rotation of a crustal block situated at the limit between a stable Europe and the microplate of the eastern part of the Carpatho–Pannonian domain. This rotation shows that the faults from Fig. 2(a) were probably still tectonically active between Sarmatian and Pannonian. The palaeomagnetic results for Pannonian and Pontian from the studied area are in good agreement with those from the Middle–Upper Miocene magmatic rocks of the Apuseni Mountains and CGH sector and show the cessation of tectonic movements detectable by the palaeomagnetic method in this part of the Carpatho–Pannonian domain during the last part of the Miocene Period.

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